

```
clear;clc

%(a)
Hour = [6;7;8;9;10;11;12;13;14;15;16;17];
SolarGen = [0.5;1.2;2.4;3.8;4.6;5.1;5.4;5.0;4.2;3.0;1.7;0.8];
Demand = [1.8;2.0;2.3;2.7;3.1;3.4;3.8;4.0;3.7;3.4;3.0;2.5];
Battery = [3.2;3.0;3.1;3.8;4.5;5.2;5.9;6.3;6.5;6.1;5.4;4.7];

%(b)
A = [Hour, SolarGen, Demand, Battery];

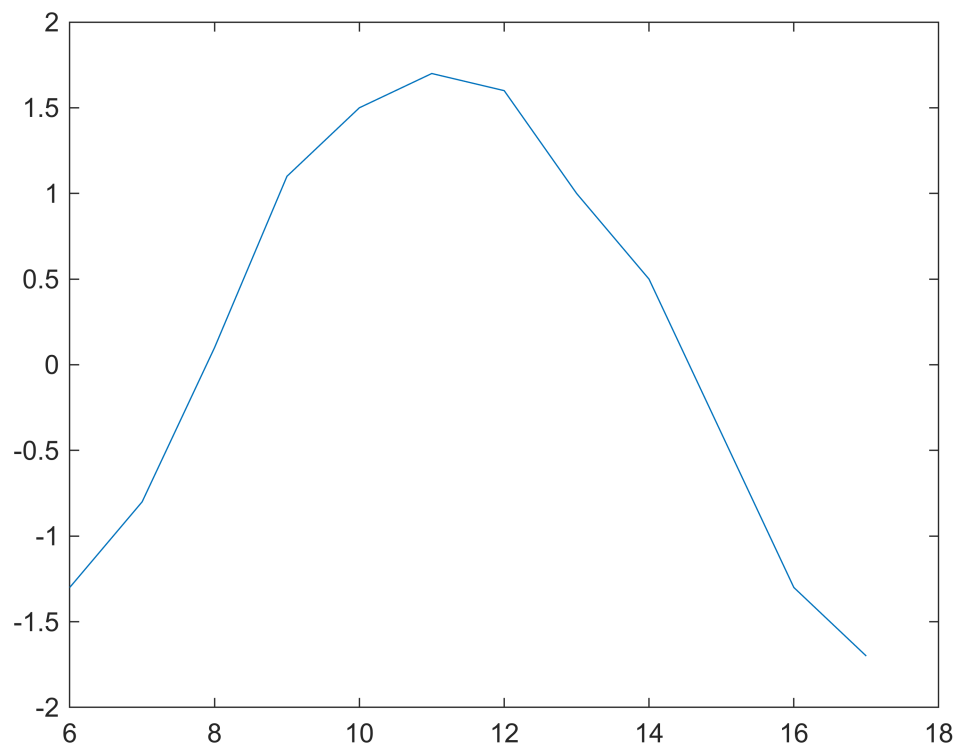
%(c)
writematrix(A, 'solar_system.txt', 'Delimiter', 'space');

%(d)
B = zeros(size(A));
B = load('solar_system.txt');

%(e)
hour = B(:,1);
solar = B(:,2);
demand = B(:,3);
battery = B(:,4);

%(f)
netPower = solar - demand;

%(g)
plot(hour,netPower);
```



2

```
clear;clc
```

```
%(a)
```

```
D = [ 1 2.1 8 12; 2 3.4 13 19; 3 1.8 7 10; 4 5.2 21 31; 5 4.7 19 28; 6 6.0 25 36; 7  
3.1 12 18; 8 5.8 24 35];
```

```
%(b)
```

```
writematrix(D,'drone_delivery.txt');  
Drone = load('drone_delivery.txt');
```

```
%(c)
```

```
RouteID = Drone(:,1);  
DistanceKm = Drone(:,2);  
TimeMin = Drone(:,3);  
BatteryUsed = Drone(:,4);
```

```
%(d)
```

```
speed = DistanceKm ./ TimeMin;
```

```
%(e)
```

```
BatteryPerKm = BatteryUsed ./ DistanceKm;
```

```
%(f)
```

```
idex = BatteryPerKm > 6;
```

```
FinalRoute = [RouteID(idex), DistanceKm(idex), BatteryPerKm(idex)];
disp(FinalRoute)
```

```
8.0000    5.8000    6.0345
```

3

```
clear;clc
```

```
%(a)
```

```
Zone = ["Z1"; "Z2"; "Z3"; "Z4"; "Z5"; "Z6"; "Z7"; "Z8"];
Temp = [22.5; 24.1; 21.8; 25.3; 23.7; 26.0; 22.9; 24.8];
Humidity = [61; 58; 67; 55; 62; 53; 65; 57];
Light = [420; 460; 390; 520; 480; 550; 410; 500];
CO2 = [510; 540; 495; 610; 575; 640; 520; 600];
```

```
%(b)
```

```
Farm = table(Zone, Temp, Humidity, Light, CO2);
```

```
%(c)
```

```
Farm.EnvIndex = 0.4*Farm.Temp + 0.01*Farm.Light + 0.02*Farm.CO2;
```

```
%(d)
```

```
index = Farm.EnvIndex >= 25;
HighEnv = Farm(index,:);
```

```
%(e)
```

```
HighEnv(:, {'Zone', 'Temp', 'Light', 'EnvIndex'})
```

```
ans = 5x4 table
```

	Zone	Temp	Light	EnvIndex
1	"Z2"	24.1000	460	25.0400
2	"Z4"	25.3000	520	27.5200
3	"Z5"	23.7000	480	25.7800
4	"Z6"	26	550	28.7000
5	"Z8"	24.8000	500	26.9200

4

```
clear;clc
```

```
%(a)
```

```
Machine = ["M1" "M2" "M3" "M4" "M5" "M6" "M7" "M8" "M9" "M10"];
Vibration = [1.8 2.4 1.5 3.1 2.7 1.9 3.5 2.2 1.6 2.9];
Temp = [68 72 65 78 74 69 82 71 66 76];
Power = [4.5 5.1 4.2 6.3 5.7 4.8 6.8 5.0 4.4 6.0];
Defect = [2 5 1 8 6 3 10 4 1 7];
Factory = table(Machine, Vibration, Temp, Power, Defect);
```

```

%(b)
Factory.RiskScore = Vibration .*Temp./Power;

%(c)
index = Factory.RiskScore >= 35 & Defect >= 6;
RiskMachines = Factory(index,:);

%(d)
RiskMachines(:, {'Machine', 'RiskScore', 'Defect'})

```

ans = 4x3 table

	Machine	RiskScore	Defect
1	"M4"	38.3810	8
2	"M5"	35.0526	6
3	"M7"	42.2059	10
4	"M10"	36.7333	7

```

%(e)
writetable(Factory,"Factory_risk.csv");

```

5

```
clear;clc
```

```

%(a)
url = 'https://archive.ics.uci.edu/ml/machine-learning-databases/abalone/abalone.data';

%(b)
Abalone = readtable(url, 'FileType', 'text', 'ReadVariableNames',false);

%(c)
Abalone.Properties.VariableNames =
{'Sex', 'Length', 'Diameter', 'Height', 'WholeWeight', 'ShuckedWeight', 'VisceraWeight',
'ShellWeight', 'Rings'};

%(d)
Abalone(:, {'WholeWeight', 'ShellWeight', 'Rings'})

```

ans = 4177x3 table

	WholeWeight	ShellWeight	Rings
1	0.5140	0.1500	15
2	0.2255	0.0700	7
3	0.6770	0.2100	9
4	0.5160	0.1550	10

	WholeWeight	ShellWeight	Rings
5	0.2050	0.0550	7
6	0.3515	0.1200	8
7	0.7775	0.3300	20
8	0.7680	0.2600	16
9	0.5095	0.1650	9
10	0.8945	0.3200	19
11	0.6065	0.2100	14
12	0.4060	0.1350	10
13	0.5415	0.1900	11
14	0.6845	0.2050	10
15	0.4755	0.1850	10
16	0.6645	0.2400	12
17	0.2905	0.1150	7
18	0.4510	0.1300	10
19	0.2555	0.1000	7
20	0.3810	0.1150	9
21	0.2455	0.0750	11
22	0.2255	0.0850	10
23	0.9395	0.2700	12
24	0.7635	0.2000	9
25	1.1615	0.3050	10
26	0.9285	0.3000	11
27	0.9955	0.2850	11
28	0.9310	0.2800	12
29	0.9365	0.2950	15
30	0.8635	0.2000	11
31	0.9975	0.3300	10
32	1.6390	0.4600	15
33	1.3380	0.3500	18
34	1.7980	0.4550	19
35	1.7095	0.4900	13
36	0.4795	0.1250	8
37	1.2170	0.3400	16

	WholeWeight	ShellWeight	Rings
38	0.5225	0.1450	8
39	0.8830	0.2600	11
40	0.3275	0.0900	9
41	0.4250	0.1150	9
42	0.8515	0.2700	14
43	0.0700	0.0200	5
44	0.0420	0.0120	5
45	0.0420	0.0150	4
46	0.2030	0.0750	7
47	0.5795	0.1400	9
48	0.4605	0.1500	7
49	0.1610	0.0450	6
50	0.8355	0.2450	9
51	0.5950	0.1900	8
52	0.3030	0.1000	7
53	0.5415	0.1600	10
54	0.4775	0.1500	10
55	0.3850	0.1100	7
56	0.6615	0.2200	8
57	0.4425	0.1350	8
58	0.5895	0.1700	8
59	0.0860	0.0250	4
60	0.5830	0.1750	7
61	0.4115	0.1350	7
62	0.6250	0.1750	9
63	0.6965	0.2000	10
64	0.3785	0.1000	7
65	0.5800	0.1850	8
66	0.4800	0.1350	8
67	0.9150	0.3200	12
68	1.2850	0.4850	13
69	0.5305	0.1700	10
70	0.1510	0.0450	6

	WholeWeight	ShellWeight	Rings
71	0.7665	0.2750	13
72	0.3530	0.1000	8
73	1.2470	0.4250	20
74	1.1850	0.3800	11
75	1.0980	0.3150	13
76	1.0075	0.2800	15
77	0.9440	0.3150	9
78	0.9220	0.3050	10
79	0.7880	0.2850	11
80	1.1025	0.3450	14
81	0.9410	0.2850	9
82	1.6150	0.6750	12
83	0.9885	0.3200	16
84	1.3175	0.5800	21
85	1.0130	0.3600	14
86	1.2950	0.4400	12
87	1.1950	0.4000	13
88	0.8645	0.2600	10
89	0.5170	0.1650	9
90	0.9775	0.3300	12
91	0.8115	0.2550	15
92	0.7550	0.2600	12
93	1.1150	0.3900	13
94	1.2620	0.3900	10
95	1.4940	0.4850	15
96	1.6060	0.4800	14
97	0.7250	0.2500	9
98	0.5230	0.1450	8
99	0.5225	0.1650	7
100	0.5785	0.1550	10

⋮

```

%(e)
index = strcmp(Abalone.Sex, 'M') & Abalone.Rings >= 15;

```

```
OldMale = Abalone(index,:);
```

```
%(f)
```

```
OldMale(:,{'Sex','Length','WholeWeight','Rings'})
```

```
ans = 167x4 table
```

	Sex	Length	WholeWeight	Rings
1	'M'	0.4550	0.5140	15
2	'M'	0.6050	0.9365	15
3	'M'	0.6650	1.3380	18
4	'M'	0.5950	1.3175	21
5	'M'	0.5650	0.8115	15
6	'M'	0.6950	1.4940	15
7	'M'	0.5500	0.8430	15
8	'M'	0.5300	0.8830	15
9	'M'	0.7000	1.7255	19
10	'M'	0.7100	1.9590	18
11	'M'	0.5950	1.2620	17
12	'M'	0.6500	1.3445	16
13	'M'	0.5900	1.0530	15
14	'M'	0.6450	1.5140	16
15	'M'	0.6400	1.3680	21
16	'M'	0.6400	1.3035	16
17	'M'	0.5600	0.9220	15
18	'M'	0.6250	1.4455	22
19	'M'	0.5650	0.9425	17
20	'M'	0.6000	1.2415	15
21	'M'	0.5600	1.0110	15
22	'M'	0.5600	1.0700	19
23	'M'	0.5900	1.1870	16
24	'M'	0.6300	1.5250	15
25	'M'	0.6550	1.8440	22
26	'M'	0.6100	1.6420	18
27	'M'	0.6350	1.4840	20
28	'M'	0.5400	0.9960	17
29	'M'	0.6100	1.1160	15
30	'M'	0.6000	1.0575	26

	Sex	Length	WholeWeight	Rings
31	'M'	0.5950	1.3305	19
32	'M'	0.6300	1.3000	23
33	'M'	0.5900	1.2830	16
34	'M'	0.6350	1.2750	16
35	'M'	0.7000	2.1300	20
36	'M'	0.7450	2.4990	17
37	'M'	0.6050	1.2270	18
38	'M'	0.6300	1.0160	19
39	'M'	0.5650	0.9950	15
40	'M'	0.6050	1.1405	15
41	'M'	0.5900	0.9110	16
42	'M'	0.5900	1.1045	15
43	'M'	0.6300	1.3520	20
44	'M'	0.6000	1.0150	19
45	'M'	0.6000	1.0360	20
46	'M'	0.5650	0.9065	18
47	'M'	0.5450	0.7505	22
48	'M'	0.5650	0.9265	16
49	'M'	0.7250	1.8910	16
50	'M'	0.6750	1.4385	21
51	'M'	0.5300	0.7395	17
52	'M'	0.6650	1.3470	17
53	'M'	0.7350	1.7560	21
54	'M'	0.6600	1.3200	16
55	'M'	0.5750	0.9325	17
56	'M'	0.5700	0.8490	15
57	'M'	0.6550	1.2635	15
58	'M'	0.6250	1.3690	17
59	'M'	0.6000	1.1050	15
60	'M'	0.5850	1.0550	15
61	'M'	0.4200	0.2885	15
62	'M'	0.4900	0.5730	21
63	'M'	0.5350	0.8345	16

	Sex	Length	WholeWeight	Rings
64	'M'	0.5650	0.9090	15
65	'M'	0.5550	1.0920	15
66	'M'	0.5250	0.9900	15
67	'M'	0.5200	0.8230	15
68	'M'	0.5250	0.8665	18
69	'M'	0.5300	0.8425	20
70	'M'	0.6950	1.5165	15
71	'M'	0.6500	1.4805	20
72	'M'	0.6150	1.1375	20
73	'M'	0.5700	1.0615	15
74	'M'	0.5850	1.2565	15
75	'M'	0.5050	0.6775	15
76	'M'	0.4650	0.5755	15
77	'M'	0.5500	1.0420	15
78	'M'	0.5150	0.6950	15
79	'M'	0.6550	2.0805	17
80	'M'	0.6600	1.3905	15
81	'M'	0.7300	2.8255	17
82	'M'	0.6400	1.5210	15
83	'M'	0.6200	1.5145	15
84	'M'	0.7750	2.0320	17
85	'M'	0.5000	0.6600	19
86	'M'	0.6650	2.1835	27
87	'M'	0.5800	1.2275	16
88	'M'	0.6200	1.8060	17
89	'M'	0.6900	1.8465	19
90	'M'	0.5350	0.9195	16
91	'M'	0.6450	1.4935	15
92	'M'	0.5700	0.9245	19
93	'M'	0.6150	1.3600	18
94	'M'	0.6600	1.6305	18
95	'M'	0.5650	1.1420	17
96	'M'	0.6300	1.5390	17

	Sex	Length	WholeWeight	Rings
97	'M'	0.5800	1.0350	17
98	'M'	0.5950	1.0965	17
99	'M'	0.5850	1.1710	17
100	'M'	0.6350	1.5980	15

⋮

6

```

%(a)
Abalone;

%(b)
Abalone.ShellRatio = Abalone.ShellWeight ./ Abalone.WholeWeight;

%(c)
index2 = Abalone.ShellRatio >= 0.25 & Abalone.Rings >= 12;
HardShell = Abalone(index2,:);

%(d)
HardShell(:,{'Sex','WholeWeight','ShellWeight','ShellRatio','Rings'})

```

ans = 896×5 table

	Sex	WholeWeight	ShellWeight	ShellRatio	Rings
1	'M'	0.5140	0.1500	0.2918	15
2	'F'	0.7775	0.3300	0.4244	20
3	'F'	0.7680	0.2600	0.3385	16
4	'F'	0.8945	0.3200	0.3577	19
5	'F'	0.6065	0.2100	0.3462	14
6	'M'	0.6645	0.2400	0.3612	12
7	'F'	0.9395	0.2700	0.2874	12
8	'M'	0.9310	0.2800	0.3008	12
9	'M'	0.9365	0.2950	0.3150	15
10	'F'	1.6390	0.4600	0.2807	15
11	'M'	1.3380	0.3500	0.2616	18
12	'F'	1.7980	0.4550	0.2531	19
13	'F'	1.7095	0.4900	0.2866	13
14	'F'	1.2170	0.3400	0.2794	16
15	'F'	0.8515	0.2700	0.3171	14
16	'F'	0.9150	0.3200	0.3497	12

	Sex	WholeWeight	ShellWeight	ShellRatio	Rings
17	'F'	1.2850	0.4850	0.3774	13
18	'M'	0.7665	0.2750	0.3588	13
19	'F'	1.2470	0.4250	0.3408	20
20	'F'	1.0980	0.3150	0.2869	13
21	'F'	1.0075	0.2800	0.2779	15
22	'F'	1.1025	0.3450	0.3129	14
23	'M'	1.6150	0.6750	0.4180	12
24	'F'	0.9885	0.3200	0.3237	16
25	'M'	1.3175	0.5800	0.4402	21
26	'M'	1.0130	0.3600	0.3554	14
27	'F'	1.2950	0.4400	0.3398	12
28	'M'	1.1950	0.4000	0.3347	13
29	'F'	0.9775	0.3300	0.3376	12
30	'M'	0.8115	0.2550	0.3142	15
31	'M'	0.7550	0.2600	0.3444	12
32	'M'	1.1150	0.3900	0.3498	13
33	'M'	1.4940	0.4850	0.3246	15
34	'M'	1.6060	0.4800	0.2989	14
35	'M'	0.8430	0.2550	0.3025	15
36	'M'	0.8830	0.3350	0.3794	15
37	'M'	1.1735	0.3450	0.2940	12
38	'F'	0.7270	0.2350	0.3232	12
39	'F'	0.7660	0.2550	0.3329	14
40	'M'	1.7255	0.5400	0.3130	19
41	'M'	1.9590	0.7800	0.3982	18
42	'M'	1.2620	0.4100	0.3249	17
43	'M'	1.3445	0.4465	0.3321	16
44	'M'	1.0530	0.3250	0.3086	15
45	'F'	0.9885	0.3100	0.3136	12
46	'F'	0.9940	0.3400	0.3421	13
47	'M'	1.5140	0.6350	0.4194	16
48	'F'	0.9075	0.2850	0.3140	13
49	'F'	1.1240	0.4400	0.3915	13

	Sex	WholeWeight	ShellWeight	ShellRatio	Rings
50	'M'	1.0560	0.3400	0.3220	13
51	'M'	1.2445	0.3450	0.2772	12
52	'F'	2.1410	1.0050	0.4694	18
53	'F'	1.7520	0.8150	0.4652	16
54	'M'	2.5500	0.7250	0.2843	14
55	'F'	2.1240	0.8500	0.4002	20
56	'F'	1.8420	0.6500	0.3529	20
57	'M'	1.9810	0.7600	0.3836	14
58	'F'	1.6185	0.6200	0.3831	12
59	'M'	1.9565	0.5900	0.3016	14
60	'M'	0.9395	0.2950	0.3140	14
61	'M'	1.3680	0.5700	0.4167	21
62	'F'	1.5675	0.4600	0.2935	12
63	'M'	1.2180	0.3550	0.2915	13
64	'F'	1.0525	0.2850	0.2708	12
65	'M'	1.1595	0.3250	0.2803	13
66	'M'	1.0225	0.3600	0.3521	12
67	'M'	0.9270	0.3600	0.3883	14
68	'M'	0.7720	0.2450	0.3174	12
69	'M'	1.3035	0.4650	0.3567	16
70	'M'	0.9220	0.2600	0.2820	15
71	'M'	0.8250	0.2850	0.3455	13
72	'F'	0.6300	0.2300	0.3651	12
73	'F'	0.8155	0.2750	0.3372	13
74	'F'	0.8325	0.3100	0.3724	13
75	'F'	0.3800	0.1200	0.3158	12
76	'F'	0.6600	0.2350	0.3561	14
77	'F'	0.7830	0.2450	0.3129	15
78	'M'	0.8360	0.2750	0.3289	13
79	'M'	0.9025	0.3250	0.3601	14
80	'M'	1.4455	0.4350	0.3009	22
81	'I'	0.4350	0.1250	0.2874	12
82	'M'	0.9425	0.2750	0.2918	17

	Sex	WholeWeight	ShellWeight	ShellRatio	Rings
83	'I'	0.1585	0.0500	0.3155	13
84	'M'	1.1235	0.3600	0.3204	13
85	'F'	1.0605	0.3000	0.2829	12
86	'F'	1.0940	0.3450	0.3154	15
87	'M'	1.2415	0.3400	0.2739	15
88	'M'	1.0110	0.3700	0.3660	15
89	'M'	1.0700	0.4100	0.3832	19
90	'F'	1.4150	0.3650	0.2580	15
91	'F'	1.1015	0.2950	0.2678	13
92	'F'	0.8650	0.2700	0.3121	12
93	'M'	0.8950	0.2350	0.2626	14
94	'F'	0.4995	0.1700	0.3403	13
95	'F'	1.7790	0.5500	0.3092	22
96	'M'	1.1870	0.3700	0.3117	16
97	'M'	1.2190	0.3850	0.3158	14
98	'M'	1.5250	0.4500	0.2951	15
99	'M'	1.2590	0.4100	0.3257	13
100	'M'	1.8440	0.5850	0.3172	22

⋮

```
%(e)
ratioData = HardShell{:, 'ShellRatio'};
```

7

```
clear; clc

%(a)
url = 'https://archive.ics.uci.edu/ml/machine-learning-databases/forest-fires/forestfires.csv';

%(b)
ForestFires = readtable(url);

%(c)
temp = ForestFires.temp;
RH = ForestFires.RH;
wind = ForestFires.wind;
rain = ForestFires.rain;
```

```
area = ForestFires.area;
```

```
%(d)
```

```
index = temp >= 25 & RH <= 35;
```

```
HotDryFires = ForestFires(index, :);
```

```
%(e)
```

```
HotDryFires(:, {'month', 'day', 'temp', 'RH', 'wind', 'area'})
```

```
ans = 54x6 table
```

	month	day	temp	RH	wind	area
1	'sep'	'wed'	26.4000	21	4.5000	0
2	'sep'	'sat'	30.2000	24	2.7000	0
3	'sep'	'fri'	25.4000	24	3.6000	0
4	'sep'	'wed'	25.9000	24	4	0
5	'sep'	'thu'	27.7000	24	2.2000	0
6	'sep'	'fri'	27.6000	30	1.3000	0
7	'sep'	'thu'	25.4000	27	2.2000	0
8	'aug'	'thu'	27.4000	22	4	0.9000
9	'sep'	'sat'	29.6000	27	2.7000	1.4600
10	'sep'	'sat'	28.6000	27	2.2000	1.6100
11	'sep'	'sun'	28.3000	26	3.1000	64.1000
12	'sep'	'wed'	26.4000	21	4.5000	88.4900
13	'sep'	'sun'	27.8000	27	3.1000	95.1800
14	'sep'	'sat'	25.1000	27	4	1.0908e+03
15	'aug'	'wed'	28.7000	28	2.7000	0
16	'aug'	'wed'	26.8000	25	3.1000	0.6800
17	'aug'	'mon'	28.3000	32	4	8.8500
18	'jul'	'tue'	26.9000	28	5.4000	86.4500
19	'sep'	'sun'	25.3000	27	2.7000	0
20	'aug'	'wed'	28.2000	29	1.8000	5.8600
21	'aug'	'sat'	25.9000	32	3.1000	0
22	'jun'	'wed'	28	34	4.5000	0
23	'jun'	'wed'	28	34	4.5000	8.1600
24	'aug'	'tue'	27.9000	27	2.2000	0
25	'aug'	'sat'	26.4000	33	3.6000	0
26	'aug'	'thu'	27.5000	27	4.9000	746.2800
27	'aug'	'wed'	30.8000	19	4.5000	0

	month	day	temp	RH	wind	area
28	'jul'	'sun'	29.3000	27	3.6000	6.3000
29	'aug'	'sat'	26.9000	31	3.6000	4.9600
30	'aug'	'mon'	27.9000	33	2.2000	2.3500
31	'aug'	'thu'	26.2000	34	5.8000	0
32	'aug'	'mon'	26.8000	35	1.3000	0.5400
33	'aug'	'mon'	25.5000	29	1.8000	1.2300
34	'jun'	'thu'	26.4000	35	2.7000	10.0800
35	'jul'	'thu'	27.2000	28	1.3000	1.7600
36	'jul'	'thu'	30.2000	25	4.5000	2.7500
37	'jul'	'thu'	30.2000	22	4.9000	0
38	'aug'	'sun'	31	27	5.4000	0
39	'aug'	'sun'	33.1000	25	4	26.4300
40	'aug'	'mon'	30.6000	28	3.6000	2.0700
41	'aug'	'tue'	26.4000	34	3.6000	16.4000
42	'aug'	'wed'	28.7000	33	4	0
43	'aug'	'thu'	32.4000	21	4.5000	0
44	'aug'	'fri'	32.4000	27	2.2000	0
45	'aug'	'fri'	27.5000	29	4.5000	43.3200
46	'aug'	'sat'	30.8000	30	4.9000	8.5900
47	'aug'	'mon'	32.6000	26	3.1000	2.7700
48	'aug'	'tue'	32.3000	27	2.2000	14.6800
49	'aug'	'tue'	33.3000	26	2.7000	40.5400
50	'aug'	'wed'	29.2000	30	4.9000	1.9500
51	'aug'	'wed'	28.9000	29	4.9000	49.5900
52	'aug'	'thu'	26.7000	35	1.8000	5.8000
53	'aug'	'sun'	27.8000	35	2.7000	0
54	'aug'	'sun'	27.8000	32	2.7000	6.4400

8

%(a)

```
Fire = ForestFires;
```

%(b)

```
Fire.RiskIndex = Fire.temp .* Fire.wind ./ Fire.RH;
```

```
%(c)
```

```
index = Fire.RiskIndex >= 2 & Fire.area > 0;  
RiskFire = Fire(index, :);
```

```
%(d)
```

```
RiskFire(:, {'month', 'day', 'temp', 'RH', 'wind', 'area', 'RiskIndex'})
```

```
ans = 105x7 table
```

	month	day	temp	RH	wind	area	RiskIndex
1	'aug'	'wed'	23.3000	31	4.5000	0.5500	3.3823
2	'aug'	'fri'	21.2000	51	8.9000	0.6100	3.6996
3	'aug'	'wed'	23.8000	32	5.4000	0.7700	4.0163
4	'aug'	'thu'	27.4000	22	4	0.9000	4.9818
5	'sep'	'tue'	24.2000	28	3.6000	0.9600	3.1114
6	'aug'	'tue'	17.4000	43	6.7000	1.0700	2.7112
7	'sep'	'thu'	23.7000	25	4.5000	1.1200	4.2660
8	'jun'	'fri'	23.2000	39	5.4000	1.1900	3.2123
9	'sep'	'fri'	20.1000	47	4.9000	1.4600	2.0955
10	'sep'	'sat'	29.6000	27	2.7000	1.4600	2.9600
11	'sep'	'sat'	28.6000	27	2.2000	1.6100	2.3304
12	'aug'	'wed'	20.5000	35	4	1.6400	2.3429
13	'sep'	'fri'	19	34	5.8000	1.6900	3.2412
14	'mar'	'sat'	15.2000	31	8.5000	1.9400	4.1677
15	'aug'	'wed'	23.4000	33	4.5000	2.5100	3.1909
16	'mar'	'fri'	14.6000	26	9.4000	2.5300	5.2785
17	'aug'	'sun'	20.1000	39	5.4000	2.7400	2.7831
18	'aug'	'sat'	20.3000	39	4.9000	4.5300	2.5505
19	'sep'	'sun'	23.9000	32	6.7000	5.3300	5.0041
20	'oct'	'mon'	19.1000	32	4	5.4400	2.3875
21	'aug'	'fri'	20.8000	34	4.9000	6.9600	2.9976
22	'aug'	'thu'	24.2000	28	2.7000	8.6800	2.3336
23	'aug'	'tue'	24.6000	22	4.5000	8.7100	5.0318
24	'sep'	'wed'	24.3000	25	4	9.4100	3.8880
25	'aug'	'tue'	24.6000	22	4.5000	10.0100	5.0318
26	'aug'	'fri'	23.5000	36	5.4000	10.0200	3.5250
27	'sep'	'mon'	22.6000	38	3.6000	11.3200	2.1411
28	'aug'	'sat'	22.9000	31	7.2000	15.4500	5.3187

	month	day	temp	RH	wind	area	RiskIndex
29	'sep'	'fri'	19.6000	33	6.3000	19.2300	3.7418
30	'sep'	'thu'	23.2000	26	4.9000	23.4100	4.3723
31	'oct'	'sat'	18.4000	25	3.1000	24.2300	2.2816
32	'sep'	'fri'	20.1000	47	4.9000	26.1300	2.0955
33	'mar'	'sat'	17	27	4.9000	28.6600	3.0852
34	'mar'	'sat'	17	27	4.9000	28.6600	3.0852
35	'sep'	'sun'	22.9000	39	4.9000	48.5500	2.8772
36	'aug'	'sat'	20.1000	34	4.5000	58.3000	2.6603
37	'sep'	'sun'	28.3000	26	3.1000	64.1000	3.3742
38	'sep'	'wed'	26.4000	21	4.5000	88.4900	5.6571
39	'sep'	'sun'	27.8000	27	3.1000	95.1800	3.1919
40	'sep'	'tue'	24.3000	36	3.1000	105.6600	2.0925
41	'sep'	'sat'	17.7000	25	3.1000	154.8800	2.1948
42	'aug'	'sun'	19.6000	41	5.8000	196.4800	2.7727
43	'sep'	'sat'	25.1000	27	4	1.0908e+03	3.7185
44	'aug'	'sun'	22.4000	54	7.6000	2.8700	3.1526
45	'aug'	'sun'	26.8000	38	6.3000	0.7600	4.4432
46	'aug'	'sun'	25.7000	39	5.4000	0.0900	3.5585
47	'aug'	'wed'	26.8000	25	3.1000	0.6800	3.3232
48	'aug'	'wed'	24	36	3.1000	0.2400	2.0667
49	'aug'	'wed'	22.1000	37	3.6000	0.2100	2.1503
50	'aug'	'mon'	28.3000	32	4	8.8500	3.5375
51	'aug'	'tue'	22.1000	54	7.6000	0.7900	3.1104
52	'jul'	'tue'	26.9000	28	5.4000	86.4500	5.1879
53	'jun'	'mon'	19.1000	39	5.4000	3.5200	2.6446
54	'sep'	'sun'	23.8000	35	3.6000	5.1800	2.4480
55	'sep'	'thu'	19.2000	24	4.9000	3.7800	3.9200
56	'sep'	'thu'	21.6000	28	6.3000	4.4100	4.8600
57	'sep'	'thu'	18.9000	34	7.2000	34.3600	4.0024
58	'sep'	'thu'	16.8000	28	4	7.2100	2.4000
59	'sep'	'thu'	16.8000	28	4	1.0100	2.4000
60	'sep'	'sat'	23.5000	27	4	3.3300	3.4815
61	'sep'	'sat'	24.2000	27	3.1000	6.5800	2.7785

	month	day	temp	RH	wind	area	RiskIndex
62	'sep'	'sat'	21.5000	28	4.5000	15.6400	3.4554
63	'sep'	'fri'	20.8000	35	4.9000	13.0600	2.9120
64	'sep'	'fri'	20.8000	35	4.9000	1.2600	2.9120
65	'sep'	'tue'	15.9000	38	5.4000	1.7500	2.2595
66	'jul'	'wed'	19.3000	39	7.2000	7.7300	3.5631
67	'aug'	'sat'	21.3000	44	4.5000	12.1800	2.1784
68	'sep'	'sun'	20.6000	55	5.4000	24.5900	2.0225
69	'aug'	'fri'	23.3000	34	3.1000	28.7400	2.1244
70	'sep'	'sun'	21.9000	43	4	70.7600	2.0372
71	'aug'	'sun'	24.3000	33	3.6000	3.6300	2.6509
72	'jun'	'wed'	28	34	4.5000	8.1600	3.7059
73	'jul'	'mon'	22.8000	27	4.5000	1.6300	3.8000
74	'aug'	'thu'	27.5000	27	4.9000	746.2800	4.9907
75	'jul'	'tue'	26.3000	39	3.1000	7.0200	2.0905
76	'aug'	'sun'	24.9000	42	5.4000	2.4400	3.2014
77	'aug'	'sun'	24.8000	36	4	3.0500	2.7556
78	'aug'	'wed'	26.2000	36	4.5000	185.7600	3.2750
79	'jul'	'sun'	29.3000	27	3.6000	6.3000	3.9067
80	'aug'	'sat'	26.9000	31	3.6000	4.9600	3.1239
81	'aug'	'sun'	24.6000	44	4	3.2000	2.2364
82	'aug'	'sat'	23.3000	40	4	6.3600	2.3300
83	'aug'	'wed'	23.4000	49	5.4000	6.4300	2.5788
84	'apr'	'sun'	13.7000	33	9.4000	61.1300	3.9024
85	'jun'	'thu'	26.4000	35	2.7000	10.0800	2.0366
86	'jun'	'thu'	22.7000	40	9.4000	3.1900	5.3345
87	'jul'	'sun'	26.1000	45	4	7.3600	2.3200
88	'jul'	'thu'	30.2000	25	4.5000	2.7500	5.4360
89	'aug'	'sun'	23.4000	40	5.8000	1.2900	3.3930
90	'aug'	'sun'	33.1000	25	4	26.4300	5.2960
91	'aug'	'mon'	30.6000	28	3.6000	2.0700	3.9343
92	'aug'	'tue'	24.1000	43	6.3000	2	3.5309
93	'aug'	'tue'	26.4000	34	3.6000	16.4000	2.7953
94	'aug'	'tue'	19.4000	71	7.6000	46.7000	2.0766

	month	day	temp	RH	wind	area	RiskIndex
95	'aug'	'fri'	27.5000	29	4.5000	43.3200	4.2672
96	'aug'	'sat'	30.8000	30	4.9000	8.5900	5.0307
97	'aug'	'mon'	32.6000	26	3.1000	2.7700	3.8869
98	'aug'	'tue'	32.3000	27	2.2000	14.6800	2.6319
99	'aug'	'tue'	33.3000	26	2.7000	40.5400	3.4581
100	'aug'	'tue'	27.3000	63	4.9000	10.8200	2.1233

⋮

```
%(e)
writetable(RiskFire, 'forest_risk.csv');
```

9

```
clear; clc
```

```
%(a)
P = readtable('penguins.csv');
```

경고: 테이블에 대한 변수 이름을 생성하기 전에 파일의 열 제목이 유효한 MATLAB 식별자가 되도록 수정되었습니다. 원래 열 제목은 `VariableDescriptions` 속성에 저장되어 있습니다.

원래 열 제목을 테이블 변수 이름으로 사용하려면 `'VariableNamingRule'`을 `'preserve'`로 설정하십시오.

```
%(b)
P.Properties.VariableNames
```

```
ans = 1×19 cell
'studyName' 'SampleNumber' 'species' 'Region' 'Latitude' 'Longitude' '...
```

```
%(c)
index = P.body_mass_g >= 4500;
HeavyPenguins = P(index, :);
```

```
%(d)
HeavyPenguins(:, {'species', 'island', 'body_mass_g', 'sex'})
```

```
ans = 118×4 table
```

	species	island	body_mass_g	sex
1	'Adelie Penguin (Pygoscelis adeliae)'	'Torgersen'	4675	'MALE'
2	'Adelie Penguin (Pygoscelis adeliae)'	'Torgersen'	4500	'MALE'
3	'Adelie Penguin (Pygoscelis adeliae)'	'Dream'	4650	'MALE'
4	'Adelie Penguin (Pygoscelis adeliae)'	'Dream'	4600	'MALE'
5	'Adelie Penguin (Pygoscelis adeliae)'	'Torgersen'	4700	'MALE'

	species	island	body_mass_g	sex
6	'Adelie Penguin (Pygoscelis adeliae)'	'Biscoe'	4725	'MALE'
7	'Adelie Penguin (Pygoscelis adeliae)'	'Biscoe'	4775	'MALE'
8	'Adelie Penguin (Pygoscelis adeliae)'	'Biscoe'	4600	'MALE'
9	'Chinstrap penguin (Pygoscelis antarctica)'	'Dream'	4550	'MALE'
10	'Chinstrap penguin (Pygoscelis antarctica)'	'Dream'	4800	'MALE'
11	'Chinstrap penguin (Pygoscelis antarctica)'	'Dream'	4500	'MALE'
12	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4500	'FEMALE'
13	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5700	'MALE'
14	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5700	'MALE'
15	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5400	'MALE'
16	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4550	'FEMALE'
17	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4800	'FEMALE'
18	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5200	'MALE'
19	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5150	'MALE'
20	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4650	'FEMALE'
21	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5550	'MALE'
22	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4650	'FEMALE'
23	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5850	'MALE'
24	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5850	'MALE'
25	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	6300	'MALE'
26	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4800	'FEMALE'
27	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5350	'MALE'
28	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5700	'MALE'
29	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5000	'FEMALE'
30	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5050	'MALE'
31	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5000	'FEMALE'
32	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5100	'MALE'
33	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5650	'MALE'
34	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4600	'FEMALE'
35	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5550	'MALE'
36	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5250	'MALE'
37	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4700	'FEMALE'
38	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5050	'FEMALE'

	species	island	body_mass_g	sex
39	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	6050	'MALE'
40	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5150	'FEMALE'
41	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5400	'MALE'
42	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4950	'FEMALE'
43	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5250	'MALE'
44	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5350	'MALE'
45	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5700	'MALE'
46	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4750	'MALE'
47	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5550	'MALE'
48	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4900	'FEMALE'
49	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5400	'MALE'
50	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5100	'FEMALE'
51	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5300	'MALE'
52	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4850	'FEMALE'
53	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5300	'MALE'
54	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5000	'MALE'
55	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4900	'FEMALE'
56	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5050	'MALE'
57	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5000	'MALE'
58	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5550	'MALE'
59	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5300	'MALE'
60	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5650	'MALE'
61	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4700	'FEMALE'
62	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5700	'MALE'
63	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4650	"
64	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5800	'MALE'
65	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4700	'FEMALE'
66	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5550	'MALE'
67	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4750	'FEMALE'
68	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5000	'MALE'
69	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5100	'MALE'
70	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5200	'FEMALE'
71	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4700	'FEMALE'

	species	island	body_mass_g	sex
72	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5800	'MALE'
73	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4600	'FEMALE'
74	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	6000	'MALE'
75	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4750	'FEMALE'
76	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5950	'MALE'
77	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4625	'FEMALE'
78	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5450	'MALE'
79	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4725	'FEMALE'
80	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5350	'MALE'
81	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4750	'FEMALE'
82	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5600	'MALE'
83	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4600	'FEMALE'
84	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5300	'MALE'
85	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4875	'FEMALE'
86	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5550	'MALE'
87	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4950	'FEMALE'
88	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5400	'MALE'
89	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4750	'FEMALE'
90	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5650	'MALE'
91	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4850	'FEMALE'
92	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5200	'MALE'
93	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4925	'MALE'
94	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4875	'FEMALE'
95	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4625	'FEMALE'
96	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5250	'MALE'
97	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4850	'FEMALE'
98	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5600	'MALE'
99	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	4975	'FEMALE'
100	'Gentoo penguin (Pygoscelis papua)'	'Biscoe'	5500	'MALE'

⋮

```

%(e)
index2 = strcmp(P.species, 'Gentoo') & P.body_mass_g >= 5000;
LargeGentoo = P(index2, :);

```

```
%(a)
```

```
P;
```

```
%(b)
```

```
P.BillRatio = P.bill_length_mm ./ P.bill_depth_mm;
```

```
%(c)
```

```
index = P.BillRatio >= 2.8;
```

```
LongBill = P(index, :);
```

```
%(d)
```

```
LongBill(:, {'species', 'bill_length_mm', 'bill_depth_mm', 'BillRatio'})
```

```
ans = 131x4 table
```

	species	bill_length_mm	bill_depth_mm	BillRatio
1	'Chinstrap penguin (Pygoscelis antarctica)'	51.3000	18.2000	2.8187
2	'Chinstrap penguin (Pygoscelis antarctica)'	52	18.1000	2.8729
3	'Chinstrap penguin (Pygoscelis antarctica)'	58	17.8000	3.2584
4	'Chinstrap penguin (Pygoscelis antarctica)'	47.5000	16.8000	2.8274
5	'Chinstrap penguin (Pygoscelis antarctica)'	46.9000	16.6000	2.8253
6	'Chinstrap penguin (Pygoscelis antarctica)'	50.9000	17.9000	2.8436
7	'Chinstrap penguin (Pygoscelis antarctica)'	49.8000	17.3000	2.8786
8	'Chinstrap penguin (Pygoscelis antarctica)'	48.1000	16.4000	2.9329
9	'Chinstrap penguin (Pygoscelis antarctica)'	46.8000	16.5000	2.8364
10	'Chinstrap penguin (Pygoscelis antarctica)'	55.8000	19.8000	2.8182
11	'Gentoo penguin (Pygoscelis papua)'	46.1000	13.2000	3.4924
12	'Gentoo penguin (Pygoscelis papua)'	50	16.3000	3.0675
13	'Gentoo penguin (Pygoscelis papua)'	48.7000	14.1000	3.4539
14	'Gentoo penguin (Pygoscelis papua)'	50	15.2000	3.2895
15	'Gentoo penguin (Pygoscelis papua)'	47.6000	14.5000	3.2828
16	'Gentoo penguin (Pygoscelis papua)'	46.5000	13.5000	3.4444
17	'Gentoo penguin (Pygoscelis papua)'	45.4000	14.6000	3.1096
18	'Gentoo penguin (Pygoscelis papua)'	46.7000	15.3000	3.0523
19	'Gentoo penguin (Pygoscelis papua)'	43.3000	13.4000	3.2313
20	'Gentoo penguin (Pygoscelis papua)'	46.8000	15.4000	3.0390
21	'Gentoo penguin (Pygoscelis papua)'	40.9000	13.7000	2.9854
22	'Gentoo penguin (Pygoscelis papua)'	49	16.1000	3.0435

	species	bill_length_mm	bill_depth_mm	BillRatio
23	'Gentoo penguin (Pygoscelis papua)'	45.5000	13.7000	3.3212
24	'Gentoo penguin (Pygoscelis papua)'	48.4000	14.6000	3.3151
25	'Gentoo penguin (Pygoscelis papua)'	45.8000	14.6000	3.1370
26	'Gentoo penguin (Pygoscelis papua)'	49.3000	15.7000	3.1401
27	'Gentoo penguin (Pygoscelis papua)'	42	13.5000	3.1111
28	'Gentoo penguin (Pygoscelis papua)'	49.2000	15.2000	3.2368
29	'Gentoo penguin (Pygoscelis papua)'	46.2000	14.5000	3.1862
30	'Gentoo penguin (Pygoscelis papua)'	48.7000	15.1000	3.2252
31	'Gentoo penguin (Pygoscelis papua)'	50.2000	14.3000	3.5105
32	'Gentoo penguin (Pygoscelis papua)'	45.1000	14.5000	3.1103
33	'Gentoo penguin (Pygoscelis papua)'	46.5000	14.5000	3.2069
34	'Gentoo penguin (Pygoscelis papua)'	46.3000	15.8000	2.9304
35	'Gentoo penguin (Pygoscelis papua)'	42.9000	13.1000	3.2748
36	'Gentoo penguin (Pygoscelis papua)'	46.1000	15.1000	3.0530
37	'Gentoo penguin (Pygoscelis papua)'	44.5000	14.3000	3.1119
38	'Gentoo penguin (Pygoscelis papua)'	47.8000	15	3.1867
39	'Gentoo penguin (Pygoscelis papua)'	48.2000	14.3000	3.3706
40	'Gentoo penguin (Pygoscelis papua)'	50	15.3000	3.2680
41	'Gentoo penguin (Pygoscelis papua)'	47.3000	15.3000	3.0915
42	'Gentoo penguin (Pygoscelis papua)'	42.8000	14.2000	3.0141
43	'Gentoo penguin (Pygoscelis papua)'	45.1000	14.5000	3.1103
44	'Gentoo penguin (Pygoscelis papua)'	59.6000	17	3.5059
45	'Gentoo penguin (Pygoscelis papua)'	49.1000	14.8000	3.3176
46	'Gentoo penguin (Pygoscelis papua)'	48.4000	16.3000	2.9693
47	'Gentoo penguin (Pygoscelis papua)'	42.6000	13.7000	3.1095
48	'Gentoo penguin (Pygoscelis papua)'	44	13.6000	3.2353
49	'Gentoo penguin (Pygoscelis papua)'	48.7000	15.7000	3.1019
50	'Gentoo penguin (Pygoscelis papua)'	42.7000	13.7000	3.1168
51	'Gentoo penguin (Pygoscelis papua)'	49.6000	16	3.1000
52	'Gentoo penguin (Pygoscelis papua)'	45.3000	13.7000	3.3066
53	'Gentoo penguin (Pygoscelis papua)'	49.6000	15	3.3067
54	'Gentoo penguin (Pygoscelis papua)'	50.5000	15.9000	3.1761
55	'Gentoo penguin (Pygoscelis papua)'	43.6000	13.9000	3.1367

	species	bill_length_mm	bill_depth_mm	BillRatio
56	'Gentoo penguin (Pygoscelis papua)'	45.5000	13.9000	3.2734
57	'Gentoo penguin (Pygoscelis papua)'	50.5000	15.9000	3.1761
58	'Gentoo penguin (Pygoscelis papua)'	44.9000	13.3000	3.3759
59	'Gentoo penguin (Pygoscelis papua)'	45.2000	15.8000	2.8608
60	'Gentoo penguin (Pygoscelis papua)'	46.6000	14.2000	3.2817
61	'Gentoo penguin (Pygoscelis papua)'	48.5000	14.1000	3.4397
62	'Gentoo penguin (Pygoscelis papua)'	45.1000	14.4000	3.1319
63	'Gentoo penguin (Pygoscelis papua)'	50.1000	15	3.3400
64	'Gentoo penguin (Pygoscelis papua)'	46.5000	14.4000	3.2292
65	'Gentoo penguin (Pygoscelis papua)'	45	15.4000	2.9221
66	'Gentoo penguin (Pygoscelis papua)'	43.8000	13.9000	3.1511
67	'Gentoo penguin (Pygoscelis papua)'	45.5000	15	3.0333
68	'Gentoo penguin (Pygoscelis papua)'	43.2000	14.5000	2.9793
69	'Gentoo penguin (Pygoscelis papua)'	50.4000	15.3000	3.2941
70	'Gentoo penguin (Pygoscelis papua)'	45.3000	13.8000	3.2826
71	'Gentoo penguin (Pygoscelis papua)'	46.2000	14.9000	3.1007
72	'Gentoo penguin (Pygoscelis papua)'	45.7000	13.9000	3.2878
73	'Gentoo penguin (Pygoscelis papua)'	54.3000	15.7000	3.4586
74	'Gentoo penguin (Pygoscelis papua)'	45.8000	14.2000	3.2254
75	'Gentoo penguin (Pygoscelis papua)'	49.8000	16.8000	2.9643
76	'Gentoo penguin (Pygoscelis papua)'	46.2000	14.4000	3.2083
77	'Gentoo penguin (Pygoscelis papua)'	49.5000	16.2000	3.0556
78	'Gentoo penguin (Pygoscelis papua)'	43.5000	14.2000	3.0634
79	'Gentoo penguin (Pygoscelis papua)'	50.7000	15	3.3800
80	'Gentoo penguin (Pygoscelis papua)'	47.7000	15	3.1800
81	'Gentoo penguin (Pygoscelis papua)'	46.4000	15.6000	2.9744
82	'Gentoo penguin (Pygoscelis papua)'	48.2000	15.6000	3.0897
83	'Gentoo penguin (Pygoscelis papua)'	46.5000	14.8000	3.1419
84	'Gentoo penguin (Pygoscelis papua)'	46.4000	15	3.0933
85	'Gentoo penguin (Pygoscelis papua)'	48.6000	16	3.0375
86	'Gentoo penguin (Pygoscelis papua)'	47.5000	14.2000	3.3451
87	'Gentoo penguin (Pygoscelis papua)'	51.1000	16.3000	3.1350
88	'Gentoo penguin (Pygoscelis papua)'	45.2000	13.8000	3.2754

	species	bill_length_mm	bill_depth_mm	BillRatio
89	'Gentoo penguin (Pygoscelis papua)'	49.1000	14.5000	3.3862
90	'Gentoo penguin (Pygoscelis papua)'	52.5000	15.6000	3.3654
91	'Gentoo penguin (Pygoscelis papua)'	47.4000	14.6000	3.2466
92	'Gentoo penguin (Pygoscelis papua)'	50	15.9000	3.1447
93	'Gentoo penguin (Pygoscelis papua)'	44.9000	13.8000	3.2536
94	'Gentoo penguin (Pygoscelis papua)'	50.8000	17.3000	2.9364
95	'Gentoo penguin (Pygoscelis papua)'	43.4000	14.4000	3.0139
96	'Gentoo penguin (Pygoscelis papua)'	51.3000	14.2000	3.6127
97	'Gentoo penguin (Pygoscelis papua)'	47.5000	14	3.3929
98	'Gentoo penguin (Pygoscelis papua)'	52.1000	17	3.0647
99	'Gentoo penguin (Pygoscelis papua)'	47.5000	15	3.1667
100	'Gentoo penguin (Pygoscelis papua)'	52.2000	17.1000	3.0526

⋮

```
%(e)
billRatioData = LongBill(:, 'BillRatio');
```

11

```
clear; clc

%(a)
H = readtable('heart_failure.csv');

%(b)
H.Properties.VariableNames
```

```
ans = 1x13 cell
'age'      'anaemia'      'creatinine_phosphokinase' 'diabetes'      'ejection_fract ...
```

```
%(c)
index = H.age >= 70 & H.ejection_fraction <= 35;
HighRisk = H(index, :);

%(d)
HighRisk(:, {'age', 'ejection_fraction', 'serum_creatinine', ...
            'serum_sodium', 'DEATH_EVENT'})
```

```
ans = 31x5 table
```

	age	ejection_fraction	serum_creatinine	serum_sodium	DEATH_EVENT
1	75	20	1.9000	130	1

	age	ejection_fraction	serum_creatinine	serum_sodium	DEATH_EVENT
2	75	15	1.2000	137	1
3	80	35	9.4000	133	1
4	70	25	1	140	1
5	75	30	1.8300	134	1
6	82	30	1.2000	132	1
7	70	20	1.8300	134	1
8	80	20	4.4000	133	1
9	95	30	2	132	1
10	70	35	2.7000	138	0
11	72	20	1.3000	136	1
12	72	25	1	140	1
13	70	25	1.2000	142	1
14	85	35	1	132	1
15	80	25	1.1000	144	0
16	72	30	1.7000	138	1
17	81	35	1.1800	137	0
18	72	25	1.7000	139	1
19	75	30	1.8300	134	1
20	70	35	0.8000	133	0
21	80	35	2.1000	134	0
22	73	30	1.1800	142	0
23	70	17	1	136	0
24	73	35	1.3000	134	0
25	73	20	1.8300	134	1
26	72	25	1.2000	134	0
27	70	35	1.1000	134	0
28	70	30	1.2000	132	0
29	70	35	1.3000	139	0
30	70	35	1.2000	132	0
31	70	35	1.1000	142	0

```

%(e)
DeathGroup = H(H.DEATH_EVENT == 1, :);
SurvivalGroup = H(H.DEATH_EVENT == 0, :);

```

12

```

%(a)
H ;

%(b)
H.RiskScore = H.age ./ H.ejection_fraction + H.serum_creatinine;

%(c)
index = H.RiskScore >= 4;
RiskScoreGroup = H(index, :);

%(d)
RiskScoreGroup(:, {'age', 'ejection_fraction', 'serum_creatinine', ...
    'RiskScore', 'DEATH_EVENT'})

```

ans = 59x5 table

	age	ejection_fraction	serum_creatinine	RiskScore	DEATH_EVENT
1	75	20	1.9000	5.6500	1
2	65	20	1.3000	4.5500	1
3	50	20	1.9000	4.4000	1
4	65	20	2.7000	5.9500	1
5	90	40	2.1000	4.3500	1
6	75	15	1.2000	6.2000	1
7	80	35	9.4000	11.6857	1
8	75	38	4	5.9737	1
9	45	14	0.8000	4.0143	1
10	75	30	1.8300	4.3300	1
11	80	38	1.9000	4.0053	1
12	58	38	5.8000	7.3263	1
13	94	38	1.8300	4.3037	1
14	85	45	3	4.8889	1
15	69	35	3.5000	5.4714	1
16	60	30	2.3000	4.3000	0
17	60	38	3	4.5789	1
18	70	20	1.8300	5.3300	1
19	80	20	4.4000	8.4000	1
20	53	20	1.4000	4.0500	1
21	60	62	6.8000	7.7677	1

	age	ejection_fraction	serum_creatinine	RiskScore	DEATH_EVENT
22	95	30	2	5.1667	1
23	70	35	2.7000	4.7000	0
24	72	20	1.3000	4.9000	1
25	60	20	2.9000	5.9000	1
26	42	15	1.3000	4.1000	1
27	70	25	1.2000	4	1
28	65	25	1.8300	4.4300	1
29	69	20	1.2000	4.6500	1
30	60	25	2.5000	4.9000	1
31	60	25	1.7000	4.1000	1
32	80	25	1.1000	4.3000	0
33	72	30	1.7000	4.1000	1
34	85	60	3.2000	4.6167	0
35	86	38	1.8300	4.0932	1
36	60	40	3.7000	5.2000	1
37	46	17	2.1000	4.8059	1
38	53	35	3.4000	4.9143	0
39	60	45	6.1000	7.4333	0
40	68	25	2.1000	4.8200	0
41	72	25	1.7000	4.5800	1
42	75	30	1.8300	4.3300	1
43	72	45	2.5000	4.1000	1
44	60	25	1.7000	4.1000	0
45	59	20	2.4000	5.3500	1
46	75	38	2.5000	4.4737	1
47	60	25	2.1000	4.5000	1
48	80	35	2.1000	4.3857	0
49	60	25	3.5000	5.9000	0
50	70	17	1	5.1176	0
51	54	70	9	9.7714	1
52	73	20	1.8300	5.4800	1
53	65	25	5	7.6000	0
54	72	25	1.2000	4.0800	0

	age	ejection_fraction	serum_creatinine	RiskScore	DEATH_EVENT
55	64	25	2.4000	4.9600	0
56	65	25	1.4000	4	1
57	55	20	1.8300	4.5800	1
58	70	40	2.7000	4.4500	0
59	42	30	3.8000	5.2000	0

```

%(e)
writetable(RiskScoreGroup, 'heart_risk_group.csv');

```

13

```

clear; clc

%(a)
B = readtable('bank-full.csv');

% 만약 열이 제대로 나뉘지 않으면 아래 코드 사용
% B = readtable('bank-full.csv', 'Delimiter', ';');

%(b)
B.Properties.VariableNames

```

```

ans = 1x17 cell
'age'      'job'      'marital'  'education' 'default'  'balance'  'h...'

```

```

%(c)
age = B.age;
balance = B.balance;
loan = B.loan;
y = B.y;

%(d)
index = age >= 40 & balance >= 2000 & string(loan) == "no";
T1 = B(index, :);

%(e)
T2 = T1(:, {'age', 'balance', 'y'});

%(f)
T2.NewFlag = double(string(T2.y) == "yes");

%(g)
Subscribed = T2(T2.NewFlag == 1, :);

%(h)

```

```
balanceVec = Subscribed{:, 'balance'};

%(i)
avgBalance = mean(balanceVec);

%(j)
writetable(Subscribed, 'bank_selected.csv');
```